

A New Earth-Like Planet? Meet GJ 3378b, Our Cosmic Neighbor!

From: Space.com Mined: 2026-07-02 3 min read

Have you ever looked up at the night sky and wondered if there are other planets like Earth out there? Well, **astronomers** just found one that's super close only 25 **light-years** away! That might sound far, but in our huge Milky Way galaxy, it's like our next-door neighbor. The planet is called GJ 3378b, and it orbits a small, cool star called a **red dwarf**. What's really exciting is that it sits in the "**habitable zone**" the perfect distance from its star where temperatures could be just right for liquid water. And where there's water, there might be life! But here's the tricky part: **red dwarfs** can blast out powerful **radiation** that could strip away a planet's **atmosphere**. Without an **atmosphere**, there's no air to breathe and no protection from space. Scientists aren't sure yet if GJ 3378b has an **atmosphere**. They found the planet by watching its star wobble the planet's gravity tugs on the star as it orbits. That wobble told them the planet is about 2.3 times heavier than Earth, so it's probably a rocky world. But because it doesn't pass in front of its star from our view, we can't use telescopes to check its **atmosphere** right now. We'll have to wait until the 2040s, when a new space telescope launches, to find out if this neighbor could be a home for life. Until then, it's a mystery waiting to be solved!

Word Treasure

astronomers -- Scientists who study stars, planets, and the whole universe.

light-years -- A very big distance -- how far light can travel in one year.

red dwarf -- A small, cool star that is much dimmer than our Sun.

habitable zone -- The area around a star where a planet's temperature might let liquid water exist.

radiation -- Dangerous energy waves that can harm living things and strip away an atmosphere.

atmosphere -- A layer of gases around a planet, like the air we breathe on Earth.

Background you need

A light-year is the distance light travels in one year -- about 6 trillion miles! The Milky Way is our home galaxy, a huge spiral filled with billions of stars. Red dwarfs are the most common type of star in the universe, but they can shoot out powerful bursts of radiation. Scientists have already found thousands of planets outside our solar system, called exoplanets.

Break it down

WHO: Astronomers (scientists who study space)

WHAT: Discovered a planet (GJ 3378b) that might have liquid water

WHERE: 25 light-years away, orbiting a red dwarf star

WHY: The planet sits in the habitable zone, but its atmosphere and the star's radiation could affect whether life could exist

Why it matters

One day, this neighbor planet might help answer the big question: Are we alone in the universe?

Test what you remember

1. What is the name of the new planet?
 - ☐ (A) GJ 3378b
 - ☐ (B) Milky Way
 - ☐ (C) Red Dwarf
 - ☐ (D) Earth
2. How far away is the planet from Earth?
 - ☐ (A) 25 light-years
 - ☐ (B) 100 miles
 - ☐ (C) 1 light-year
 - ☐ (D) 25 miles
3. What kind of star does the planet orbit?
 - ☐ (A) A red dwarf
 - ☐ (B) The Sun
 - ☐ (C) A blue giant
 - ☐ (D) A white dwarf
4. What makes the habitable zone special?
 - ☐ (A) Temperatures could be right for liquid water
 - ☐ (B) It has lots of volcanoes
 - ☐ (C) It is very cold
 - ☐ (D) There is no radiation
5. What can red dwarfs blast out that could harm a planet?
 - ☐ (A) Powerful radiation
 - ☐ (B) Liquid water
 - ☐ (C) Oxygen
 - ☐ (D) Gravity
6. How did scientists find the planet?
 - ☐ (A) By watching its star wobble
 - ☐ (B) By seeing it with a telescope
 - ☐ (C) By listening to radio signals
 - ☐ (D) By visiting it

Answer key (try the quiz first!): 1: A 2: A 3: A 4: A 5: A 6: A

Questions to think about

- 1. Exciting hope:** This planet is so close that future space missions might be able to study it for signs of life. It makes people dream of finding another Earth-like world.
- 2. Caution:** Red dwarfs are stormy and could strip away the planet's atmosphere with radiation, making it impossible for life to survive there. We just don't know yet.
- 3. Scientific patience:** We'll need to wait until the 2040s for a new space telescope to check the atmosphere. Until then, it's a mystery that science is still solving step by step.

MY THOUGHTS (at least 20 words):